

ANALYTIC GEOMETRY

CONIC SECTION

The conic sections are the curves generated by the intersection of a plane with one or two *nappes* right circular cones

Every conic section has certain features, including at least one focus and directrix. Geometrically, it can be defined as the set of points *P* whose distance to the focus is a constant multiple of the distance from *P* to the directrix of the conic.

There are four primary conic sections - the circle, the parabola, the ellipse, and the hyperbola.

General Equation of Conic Sections

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

- * where A, B, C, D, E, and F are real numbers
- * A and C cannot be both 0

CIRCLE

✧ A circle is defined as the set of all points that are at the same distance from a given point in the plane. The circle is named after its center.

- **Center:** fixed point
- **Radius:** a segment whose endpoints are the center and a point on the circle
- **Chord:** a segment whose endpoints are points on the circle
- **Diameter:** a chord passing through the center of the circle (longest chord, twice the radius)
- **Eccentricity:** 0

Standard Form of Circle

$$(x - h)^2 + (y - k)^2 = r^2$$

- * (h, k) = center and r = radius of the circle
- * $r = \sqrt{r^2}$

General Form of Circle

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

✧ Completing a Square

1. Half the coefficient of x
2. Square it
3. Add it to the expression

Example: $x^2 + 6x$

1. $6/2 = 3$
2. $3^2 = 9$
3. $x^2 + 6x + 9$

✧ Squaring a Binomial

1. Square the first term
2. Twice the product of the two terms
3. Square the last term

Example: $(y - 3)^2$

1. y^2
2. $2(-3)(y) = -6y$
3. $(-3)^2 = 9$
4. Answer = $y^2 - 6y + 9$

✧ Factoring a Binomial (Perfect Square)

1. Get the square root of first term
2. Get the square root of last term
3. Join the square roots with the sign of the middle term, and square it

Example: $y^2 - 8y + 16$

1. $\sqrt{y^2} = y$
2. $\sqrt{16} = 4$
3. $(y - 4)^2$

✧ Transform the circle in general form

1. Square the binomial
2. Transform the constant term to the left side
3. Arrange the terms and add similar terms

Example: $(x + 4)^2 + (y - 5)^2 = 36$

1. $(x + 4)^2 = x^2 + 8x + 16$
 $(y - 5)^2 = y^2 - 10y + 25$
2. $x^2 + y^2 + 8x - 10y + 25 + 16 - 36 = 0$
3. $x^2 + y^2 + 8x - 10y + 5 = 0$

✧ Transform the circle in standard form

1. Set A = C = 1
2. Transpose the constant term to the right side
3. Group terms with same variable
4. Complete the squares

5. Use the Addition Property of Equality
6. Simplify
7. Factor the trinomials

Example: $x^2 + y^2 + 4x - 6y - 3 = 0$

1. Set $A = C = 1$
2. $x^2 + y^2 + 4x - 6y = 3$
3. $x^2 + 4x + y^2 - 6y = 3$
4. $x^2 + 4x = x^2 + 4x + 4$
 $y^2 - 6y = y^2 - 6y + 9$
5. $x^2 + 4x + 4 + y^2 - 6y + 9 = 3 + 4 + 9$
6. $x^2 + 4x + 4 + y^2 - 6y + 9 = 16$
7. $(x + 2)^2 + (y - 3)^2 = 16$

General Form to Standard Form of Circle

Center (h, k): $h = -\left(\frac{D}{2A}\right)$ and $k = -\left(\frac{E}{2A}\right)$

$$r^2: \quad r^2 = \frac{D^2 + E^2 - 4AF}{4A^2}$$

PARABOLA

A parabola is the set of all points which are equally distanced from a fixed line and a fixed point.

- Vertex at (0, 0): When the vertex is at the origin and the axis of symmetry is along the x-axis, the standard form can be used to graph the parabola.
- Eccentricity = 1

Parts of Parabola

○ **Concavity** - direction of opening of parabola.

○ **Vertex** - turning point of parabola

○ **Opening of parabola**

$$(x - h)^2 = 4a (y - k)$$

- ★ Upward if $a > 0$ (positive)
- ★ Downward if $a < 0$ (negative)

$$(y - k)^2 = 4a (x - h)$$

- ★ To the right if $a > 0$ (positive)
- ★ To the left if $a < 0$ (negative)

○ **Axis of Symmetry** - divides the parabola into two equal parts

- ★ **Vertical line:** $x = c$ (look at x-axis)
- ★ **Horizontal line:** $y = c$ (look at y-axis)

○ **Focus** - fixed point. $|a|$ from the vertex (inside the parabola)

○ **Directrix** - fixed line

$y = c$ (horizontal line) and $x = c$ (vertical line)
 $|a|$ from the vertex (outside the parabola)

Focus and directrix are equidistant from the vertex (a units from vertex).

Standard Form of Parabola

$$(x - h)^2 = 4a (y - k)$$

$$(y - k)^2 = 4a (x - h)$$

- ★ **(h, k)** - vertex of the parabola
- ★ **$|a|$** - distance of focus from the vertex or distance of directrix from the vertex

General Form of Parabola

$$Ax^2 + Dx + Ey + F = 0$$

$$By^2 + Dx + Ey + F = 0$$

◇ **Transform the parabola in general form**

1. Square the left side
2. Distribute on the right side
3. Transpose the terms to the left side
4. Simplify

Example: Transform the parabola $(y - 5)^2 = -6(x + 2)$ in general form.

1. $y^2 - 10y + 25 = -6(x + 2)$
2. $y^2 - 10y + 25 = -6x - 12$
3. $y^2 - 10y + 25 + 6x + 12 = 0$
4. $y^2 + 6x - 10y + 37 = 0$

- 